

| Assessment | Test | Domain  | Skill                            | Difficulty                               |
|------------|------|---------|----------------------------------|------------------------------------------|
| SAT        | Math | Algebra | Linear equations in one variable | Hard <div></div> <div></div> <div></div> |

Question

$-49x = -98x$

How many solutions does the given equation have?

Answer

- A. Zero
- B. Exactly one
- C. Exactly two
- D. Infinitely many

Correct Answer: B

Rationale

Choice B is correct. Adding  $98x$  to each side of the given equation yields  $49x = 0$ . Dividing each side of this equation by  $49$  yields  $x = 0$ . This means that  $0$  is the only solution to the given equation. Therefore, the given equation has exactly one solution.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.